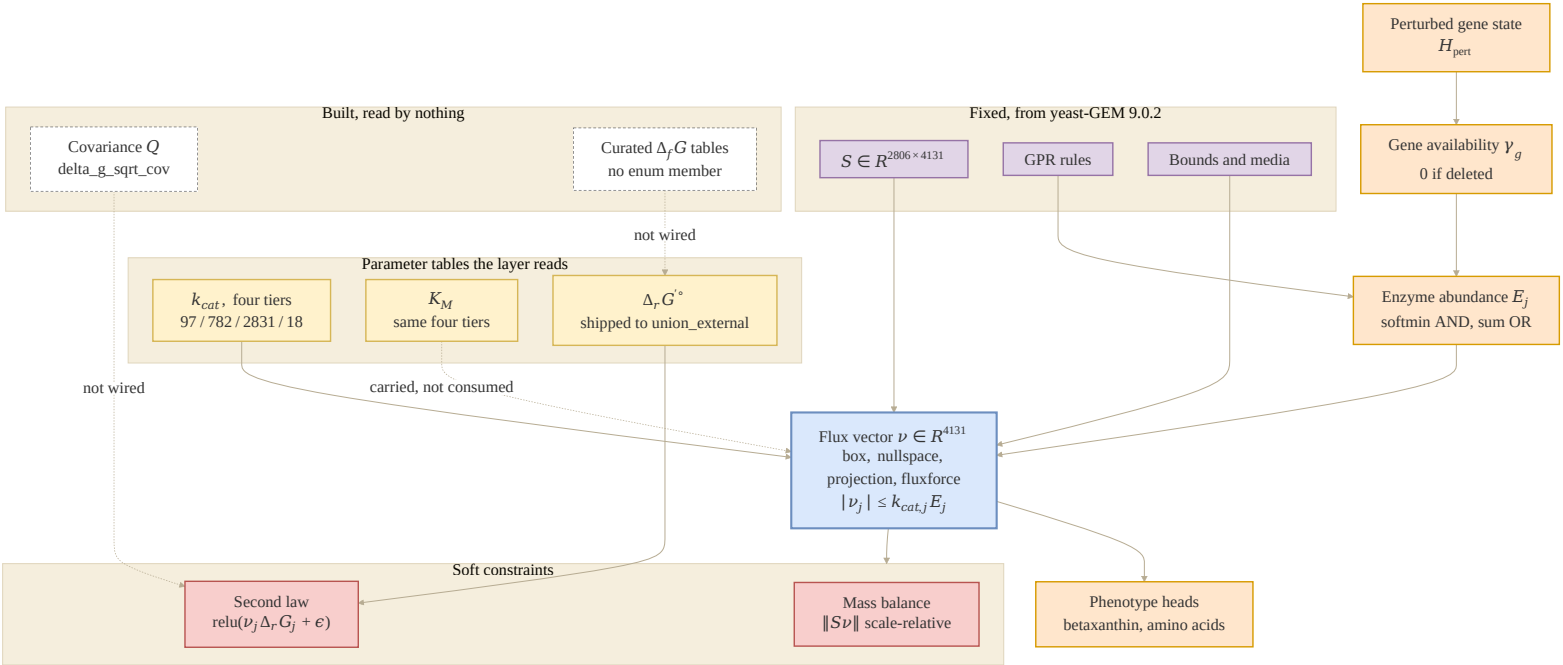

Perturbed gene state
H_{pert}
Built, read by nothing
Covariance Q
delta_g_sqrt_cov
Curated $\Delta_f G$ tables
no enum member
Fixed, from yeast-GEM 9.0.2
$S \in R^{2806 \times 4131}$
GPR rules
Bounds and media
Gene availability γ_g
0 if deleted
not wired
Parameter tables the layer reads
k_{cat}, four tiers
97 / 782 / 2831 / 18
K_M
same four tiers
$\Delta_r G'^{\circ}$
shipped to union_external
Enzyme abundance E_j
softmin AND, sum OR
not wired
carried, not consumed
Flux vector $\nu \in R^{4131}$
box, nullspace,
projection, fluxforce
$|\nu_j| \le k_{cat,j} E_j$
Soft constraints
Second law
relu($\nu_j \Delta_r G_j + \epsilon$)
Mass balance
$\|S\nu\|$ scale-relative
Phenotype heads
betaxanthin, amino acids